

SIMATS ENGINEERING
Department of Computer Science & Engineering
ASSESSMENT TOOL 2- COMPARATIVE ANALYSIS

Course Code:	CSA12	Course Title:	Computer Architecture
CO Assessed:	CO4	Assessment:	ASSESSMENT TOOL2- COMPARATIVE ANALYSIS
Weightage:	30%	Total Marks:	25
CO4 Statement:	Implement hierarchical memory systems by differentiating cache levels (L1, L2, L3), virtual memory with paging, and advanced memory technologies (DRAM, SRAM, NAND Flash, MRAM, RRAM) based on latency, bandwidth, and throughput characteristics.		
Student Name:	DIWAKAR S	Reg.No.:	192525209
Department:	AI&ML	Date:	24/8/2020

ANNEXURE A
COMPARATIVE ANALYSIS

1. Compare L1, L2, and L3 cache levels in terms of latency, bandwidth, and throughput, and analyze their impact on processor performance.
2. Differentiate SRAM and DRAM by evaluating their latency, bandwidth, and throughput characteristics in cache and main memory design.
3. Compare virtual memory with paging and cache memory mechanisms in terms of access latency and system throughput.
4. Analyze the differences between NAND Flash and DRAM with respect to latency, bandwidth, and suitability for hierarchical memory systems.
5. Compare MRAM and RRAM technologies by examining their performance in terms of access time, throughput, and energy efficiency.

Code of Conduct:

I, Diwakar S (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Diwakar S
Signature of the student:

MARKS ALLOCATION

	Scope of Items (20%)	Comparison Depth (25%)	Use of Evidence (20%)	Critical Thinking (20%)	Clarity of Presentation (15%)
Q1	0.5	1.25	1	1	0.75
Q2	0.5	1	0.5	1	0.75
Q3	1	1.25	0.5	0.5	0.5
Q4	1	1	1	0.5	0.5
Q5	0.5	1.25	1	1	0.5

20.25

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

ASSESSMENT TOOL 2- COMPARATIVE ANALYSIS

ANNEXURE A

COMPARATIVE ANALYSIS

1. Compare L1, L2, and L3 cache levels in terms of latency, bandwidth, and throughput, and analyze their impact on processor performance.

Cache memory is a small and high-speed memory located close to the CPU. It stores frequently used instructions and data so that the processor does not need to access slower main memory frequently. Modern processors generally use three cache levels: L1, L2, and L3.

L1 Cache: L1 is the smallest and fastest cache. It is normally divided into L1 instruction cache and L1 data cache. Because it is very close to the CPU execution units, it has extremely low access latency and high bandwidth. It helps the processor obtain frequently required instructions and data quickly.

L2

Cache: L2 is larger than L1 but slightly slower. It acts as a backup when the required data is not found in L1. Its larger capacity reduces the number of accesses to slower memory.

Cache: L3 is generally much larger and is often shared between multiple CPU cores. It has higher latency than L1 and L2 but provides a large pool of fast memory before the processor needs to access RAM.

Impact: The cache hierarchy improves processor performance by reducing the average memory access time. If data is found in L1, the CPU can continue execution almost immediately. If there is an L1 miss, the processor checks L2 and then L3. Only when the data is absent from all cache levels does it need to access RAM.

Therefore:

L1 → Lowest latency → Highest immediate performance
L2 → Larger capacity → Good balance of speed and size
L3 → Large shared capacity → Reduces expensive RAM accesses

A well-designed cache hierarchy increases CPU throughput, reduces processor waiting time, and improves overall system performance.

2. Differentiate SRAM and DRAM by evaluating their latency, bandwidth, and throughput characteristics in cache and main memory design.

SRAM and DRAM are two important types of semiconductor memory. They differ mainly in their structure, speed, cost, density, and power requirements.

SRAM

SRAM stores each bit using a stable circuit, typically a group of transistors. It does not require periodic refreshing while power is supplied.

Its major advantage is very low latency. Therefore, SRAM is suitable for L1, L2, and sometimes other CPU cache structures.

Because SRAM is fast, the processor can access cache data quickly. However, SRAM requires more transistors per bit, making it larger and more expensive.

DRAM

DRAM stores information using a capacitor and transistor arrangement. The stored charge gradually disappears, so DRAM requires periodic refresh operations.

DRAM is slower than SRAM but provides much greater memory density at a lower cost. Therefore, it is commonly used as main memory.

Effect on system throughput

SRAM improves processor throughput by allowing frequently accessed information to be retrieved quickly. DRAM provides large memory capacity and can support high data-transfer rates, but its higher latency means that the CPU may wait longer when data is not available in cache.

Therefore, a computer uses both:

CPU → SRAM Cache → DRAM Main Memory

This combination provides a good balance between speed, capacity, cost, and throughput.

3. Compare virtual memory with paging and cache memory mechanisms in terms of access latency and system throughput.

Virtual Memory

Virtual memory allows a system to run programs whose total virtual address space is larger than the available physical RAM. It uses secondary storage as an extension of memory.

For example, if RAM is insufficient, some memory pages may be moved to storage.

The disadvantage is that storage access is much slower than RAM. Excessive movement between RAM and storage can cause thrashing, significantly reducing system throughput.

Paging

Paging divides virtual memory into fixed-size pages and physical memory into corresponding frames.

When a program accesses a page that is not currently in RAM, a page fault occurs. The operating system must bring that page from secondary storage into RAM.

Thus, page faults have a large latency penalty.

Cache Memory

Cache memory operates differently. It stores frequently accessed data and instructions close to the processor.

When the CPU finds the required information in cache, it is called a cache hit, resulting in very low latency. If it is not found, a cache miss occurs and the CPU searches a lower memory level.

Overall impact

Cache memory primarily improves CPU performance and throughput, while virtual memory and paging primarily provide memory capacity and protection.

The hierarchy can be represented as:

CPU → Cache → RAM → Secondary Storage

Therefore, cache reduces access latency, while virtual memory and paging allow efficient management of large programs but can introduce significant delays when page faults occur.

4. Analyze the differences between NAND Flash and DRAM with respect to latency, bandwidth, and suitability for hierarchical memory systems.

NAND Flash and DRAM are both semiconductor memory technologies, but they are designed for different purposes.

NAND Flash

NAND Flash is non-volatile, meaning it can retain information even when the power is switched off.

It has high storage density and relatively low cost per bit. Therefore, it is widely used in SSDs, USB drives, memory cards, and embedded storage.

However, NAND Flash has much higher access latency than DRAM. It is therefore unsuitable as a direct replacement for CPU cache or normal main memory.

DRAM

DRAM is volatile memory and is widely used as the computer's main memory.

It provides much lower latency than NAND Flash and is designed to support frequent read and write operations during program execution.

However, DRAM requires continuous power and periodic refreshing.

Suitability in memory hierarchy

A suitable hierarchy is:

CPU Registers → SRAM Cache → DRAM → NAND Flash Storage

Each level provides a different balance between speed, capacity, and cost.

SRAM: Extremely fast but expensive.

DRAM: Fast and suitable for active programs.

NAND Flash: High capacity and persistent but slower.

Therefore, NAND Flash is mainly suitable for secondary storage, whereas DRAM is suitable for main memory.

5. Compare MRAM and RRAM technologies by examining their performance in terms of access time, throughput, and energy efficiency.

MRAM and RRAM are emerging non-volatile memory technologies designed to provide fast access while retaining data without power.

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MRAM

MRAM stores information using magnetic states rather than electrical charge.

One important advantage is that MRAM is non-volatile, so data can remain stored even when power is removed.

MRAM can provide fast access and high endurance. It can therefore be useful in embedded systems and memory applications where both speed and persistence are required.

Its energy consumption depends on the specific MRAM technology and write mechanism.

RRAM

RRAM stores data by changing the electrical resistance of a memory device.

It has the potential for high density, low-power operation, and fast switching. These properties make RRAM attractive for future memory systems and some AI-related hardware applications.

However, RRAM characteristics such as endurance, variability, and switching behavior depend strongly on the particular device technology.

Performance comparison

Access time: Both MRAM and RRAM can provide fast access compared with conventional non-volatile storage technologies.

Throughput: Both technologies can support high-throughput memory architectures, although actual performance depends on device design, memory organization, controller architecture, and workload.

Energy efficiency: Both can reduce the energy associated with maintaining stored information because they are non-volatile. RRAM is particularly attractive for low-power and high-density designs, while MRAM offers a strong combination of speed and endurance.

Conclusion

MRAM and RRAM are promising technologies for future hierarchical memory systems. MRAM is attractive when high speed, endurance, and non-volatility are important, while RRAM is attractive for high-density and potentially low-power memory applications.

Signature of the student:

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Student Name: | DIWAKAR S | Reg.No.: | 192525209

Department: | AIML | Date: | 24.8.2026

Feature | L1 Cache | L2 Cache | L3 Cache

Location | Inside CPU core | Near/inside CPU core | Shared among cores

Size | Smallest | Larger than L1 | Largest

Latency | Lowest | Low | Higher than L1/L2

Bandwidth | Very high | High | Moderate to high

Speed | Fastest | Very fast | Slower than L1/L2

Typical use | Immediate CPU data/instructions | Frequently used data | Shared frequently used data

Access | Usually per core | Usually per core | Usually shared

Feature | SRAM | DRAM

Full form | Static Random Access Memory | Dynamic Random Access Memory

Speed | Very fast | Slower than SRAM

Latency | Very low | Higher

Bandwidth | High | High, depending on memory system

Throughput | High for low-latency access | High for large sequential transfers

Refresh | Not required | Required

Density | Lower | Higher

Cost | Expensive | Less expensive

Main application | CPU cache | Main memory/RAM

Feature | Virtual Memory | Paging | Cache Memory

Main purpose | Extends usable memory using storage | Manages virtual/physical memory in pages | Speeds up CPU memory access

Main storage | RAM + secondary storage | RAM + secondary storage | SRAM

Speed | Relatively slow when storage is involved | Slower during page faults | Very fast

Latency | High when page fault occurs | High during page fault | Very low on cache hit

Unit | Virtual address space | Fixed-size pages | Cache blocks/lines

Effect on throughput | Can reduce throughput during heavy swapping | Page faults reduce throughput | Usually increases throughput

FEARTURE | NAND Flash | DRAM

Volatility | Non-volatile | Volatile

Latency | Much higher | Much lower

Bandwidth | Good for storage operations | High for main-memory access

Data retention | Retains data without power | Loses data without power

Refresh | Not required | Required

Density | Very high | Lower than NAND Flash

Cost per bit | Low | Higher

Main use | SSDs, memory cards, storage | Main memory

CPU direct working memory | Not normally suitable | Suitable

Feature | MRAM | RRAM

Full form | Magnetoresistive Random Access Memory | Resistive Random Access Memory

Data storage principle | Magnetic state | Resistance state

Volatility | Non-volatile | Non-volatile

Access time | Fast | Fast

Read operation | Very fast | Fast

Write operation | Depends on MRAM type | Depends on technology

Throughput | High | Potentially high

Energy efficiency | Good | Potentially very good

Endurance | Generally high | Technology-dependent

Applications | Embedded memory, caches, storage | Embedded memory, storage, AI hardware

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Q1 |||||

Q2 |||||

Q3 |||||

Q4 |||||

Q5 |||||

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